
London Fire Brigade: Predictive Modeling for Average Attendance Time



LONDON FIRE BRIGADE

1. Introduction and Project Scope

1.1. Project Context and Objectives

This report details the comprehensive end-to-end machine learning pipeline developed for the "London Fire Brigade Response Time" capstone project, executed as part of the Data Scientist training program at Liora. The primary objective is to accurately predict the **Average_AttendanceTimeSeconds** for LFB incidents, transforming raw operational logs into a robust predictive model.

The project adheres to a strict, milestone-driven methodology designed to demonstrate proficiency in data exploration, feature engineering, advanced machine learning, model interpretability, and application deployment.

1.2. Project Methodology and Milestone Tracking

The execution of this project was divided into structured phases, ensuring continuous validation of the data and modeling approaches prior to final deployment.

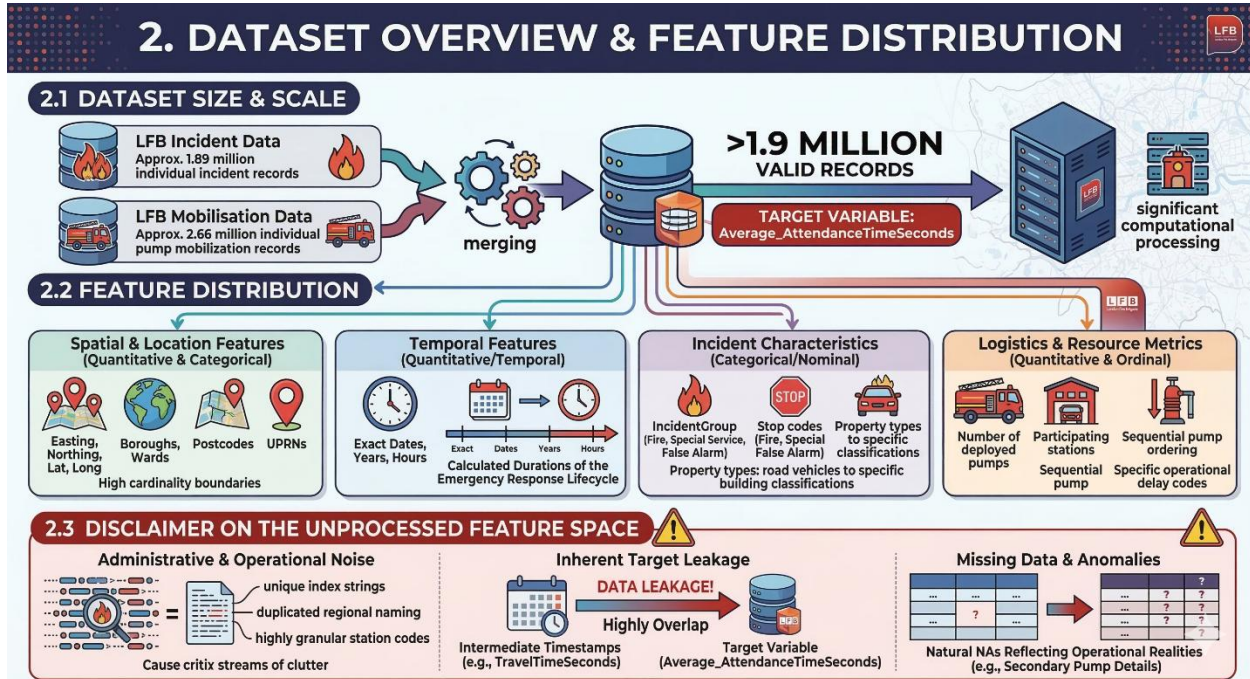
- **Phase 1: Scoping and Data Exploration**
 - Defined the project context and key operational concepts regarding LFB response times.

- Conducted an exhaustive exploratory data analysis (EDA) to highlight dataset structure, challenges, and inherent biases.
- Developed business-oriented visual analytics, validated through statistical testing.
- **Phase 2: Pre-processing and Feature Engineering**
 - Executed rigorous data cleansing and handling of missing values.
 - Engineered critical spatial and logistical features (e.g., dispatch distances) to enrich the dataset and prepare it for advanced algorithmic modeling.
- **Phase 3: Iterative Modeling Pipeline**
 - **Baseline Establishment:** Deployed initial regularized linear baseline models to evaluate raw predictive capability.
 - **Optimization:** Applied advanced statistical metrics and hyperparameter tuning to optimize performance.
 - **Advanced Modeling:** Implemented high-performance boosting algorithms (XGBoost, LightGBM) and leveraged interpretability tools to derive scientific and business-oriented conclusions.

1.3. Final Deliverables and Deployment Strategy

The culmination of this capstone project extends beyond this comprehensive written report. To ensure the predictive model is accessible and actionable, the complete codebase has been version-controlled and documented in a dedicated GitHub repository.

2. Dataset Overview and Feature Distribution



2.1. Dataset Size and Scale

The foundation of this predictive modeling project is built upon historical, real-world operational data provided by the London Fire Brigade (LFB). The scale of this dataset is massive, requiring significant computational processing. It is formed by joining two distinct operational logs:

- **LFB Incident Data:** Comprising approximately 1.89 million individual incident records.
- **LFB Mobilisation Data:** Comprising approximately 2.66 million individual pump mobilization records.

Upon merging these datasets to evaluate the overall response, the combined raw dataset yields over 1.9 million valid records containing the target variable (Average_AttendanceTimeSeconds).

2.2. Feature Distribution

The raw dataset contains a highly diverse set of features that span multiple data types and operational domains. Before dimensionality reduction, the feature space is distributed across the following core categories:

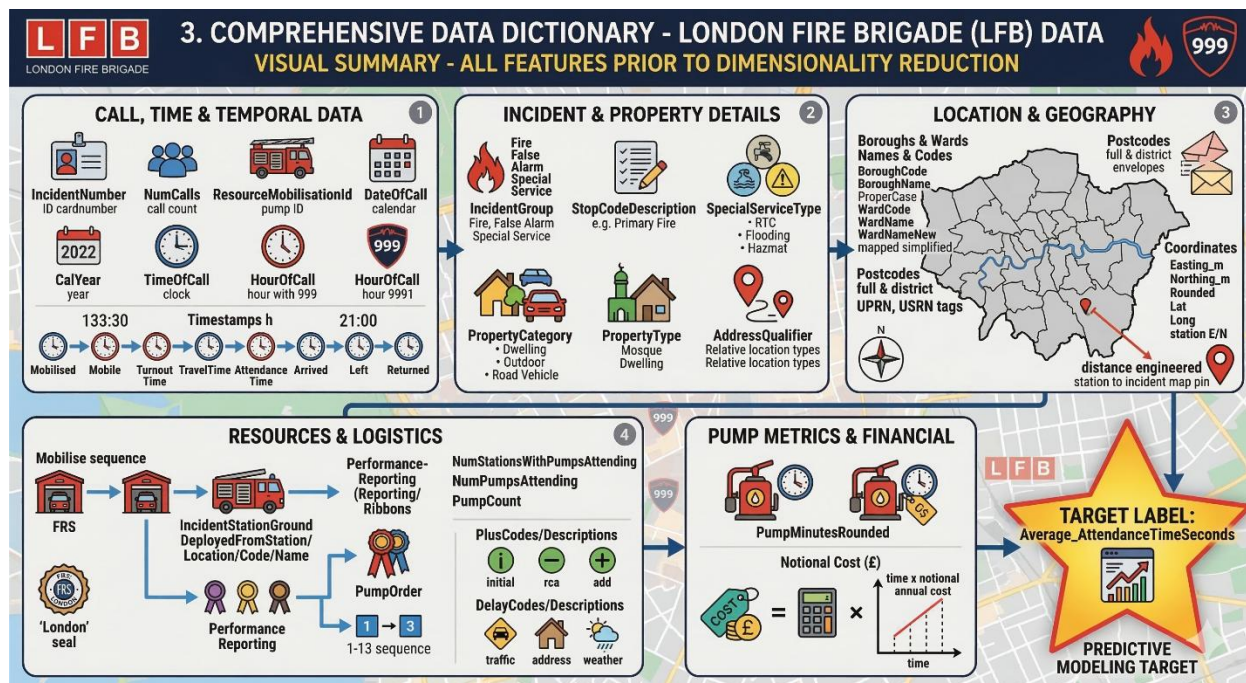
- **Spatial and Location Features (Quantitative & Categorical):** Precise spatial coordinates (Easting, Northing, Latitude, Longitude) and administrative boundaries with high cardinality (Boroughs, Wards, Postcodes, and Unique Property Reference Numbers).
- **Temporal Features (Quantitative/Temporal):** Variables tracking the exact dates, years, hours, and calculated durations of the emergency response lifecycle.
- **Incident Characteristics (Categorical/Nominal):** Descriptive features detailing the nature of the emergency, such as IncidentGroup (Fire, Special Service, False Alarm), specific stop codes, and property types ranging from road vehicles to specific building classifications.
- **Logistics and Resource Metrics (Quantitative & Ordinal):** Features tracking the physical supply-side of the response, including the number of deployed pumps, participating stations, sequential pump ordering, and specific operational delay codes.

2.3. Disclaimer on the Unprocessed Feature Space

The comprehensive data dictionary provided in Section 3 outlines the complete inventory of features as they existed in their **initial, unprocessed state**, prior to dimensionality reduction and feature engineering. When reviewing this initial dataset, several critical caveats must be noted:

1. **Administrative and Operational Noise:** The raw data contains administrative artifacts (e.g., unique index strings), duplicated regional naming conventions, and highly granular station codes that introduce significant noise.
2. **Inherent Target Leakage:** Because the data tracks the entire lifecycle of an incident, several variables mathematically overlap with the target variable (Average_AttendanceTimeSeconds). Intermediate timestamps (like TravelTimeSeconds) perfectly correlate with the target and cannot be used without causing severe data leakage.
3. **Missing Data and Anomalies:** Certain logistical features contain natural missing values (NAs) that reflect operational realities (e.g., secondary pump details when only one pump was required)

3. Comprehensive Data Dictionary



The following table provides the formal description and operational context for every feature originally present in the combined LFB Incident and Mobilisation datasets, prior to the dimensionality reduction phase.

Feature Name	Description & Operational Context
IncidentNumber	Unique ID per incident. The format changed starting from 17-Nov-2015 (9 digits before this date, and 6 digits followed by a dash and the date afterward).
DateOfCall	The date at which the 999 incident call was received (Format: yyyy-mm-dd).
CalYear	The year at which the 999 incident call was received (Format: yyyy).
TimeOfCall	The exact time at which the 999 incident call was received.

HourOfCall	The specific hour of the day at which the 999 incident call was received (Format: hh).
IncidentGroup	High-level incident category identifying the general incident type (3 groups: False Alarm, Fire, Special Service).
StopCodeDescription	Detailed incident category providing specifics across 12 categories (e.g., Primary Fire, Chimney Fire, False alarm - Malicious).
SpecialServiceType	Further detail for "Special Service" incidents across 21 categories (e.g., RTC, Flooding, Lift Release, Hazardous Materials incident).
PropertyCategory	High-level property descriptor indicating the general location type (9 groups: Road Vehicle, Outdoor, Dwelling, Non Residential, etc.).
PropertyType	Detailed property descriptor outlining 292 specific location types (e.g., Mosque).
AddressQualifier	Qualifies the specific location of the actual incident relative to the property category (11 groups).
Postcode_full	The full postal code of the incident location (redacted for Dwellings).
Postcode_district	The district-level postal code of the incident location.
UPRN	Unique Property Reference Number.
USRN	Unique Street Reference Number.
IncGeo_BoroughCode	The official administrative Borough Code for the incident location.

IncGeo_BoroughName	The official administrative Borough Name for the incident location.
ProperCase	The Borough Name formatted in proper case.
IncGeo_WardCode	The official administrative Ward Code for the incident location.
IncGeo_WardName	The official administrative Ward Name for the incident location.
IncGeo_WardNameNew	The updated/new Ward Name for the incident location.
Easting_m	The precise Easting spatial coordinate of the incident.
Northing_m	The precise Northing spatial coordinate of the incident.
Easting_rounded	The Easting spatial coordinate rounded up to the nearest 50.
Northing_rounded	The Northing spatial coordinate rounded up to the nearest 50.
Latitude	The geographical latitude of the incident location.
Longitude	The geographical longitude of the incident location.
FRS	The Fire Service ground identifier (uniformly filled as "London").
IncidentStationGround	The name of the LFB station ground responsible for the incident area.
FirstPumpArriving_AttendanceTime	The time taken in seconds for the first pump to reach the incident location.
FirstPumpArriving_DeployedFromStation	The name of the specific station from which the first pump was deployed.

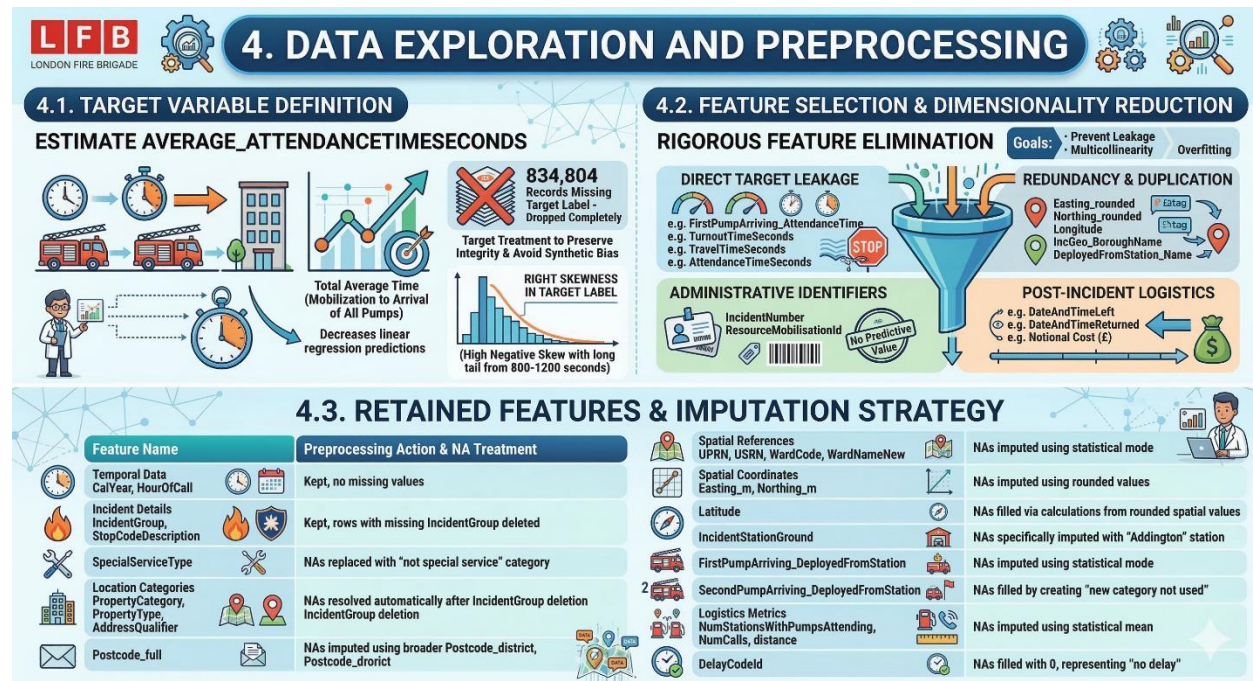
SecondPumpArriving_AttendanceTime	The time taken in seconds for the second pump to reach the incident location.
SecondPumpArriving_DeployedFromStation	The name of the specific station from which the second pump was deployed.
NumStationsWithPumpsAttending	The total count of different stations that participated in the incident response.
NumPumpsAttending	The total count of pumps that attended the incident.
PumpCount	The actual counted number of pumps deployed to the incident.
PumpMinutesRounded	The total time spent at the incident by pumps, rounded up to 60 if it was less than an hour.
Notional Cost (£)	The financial representation of the response, calculated by multiplying time spent by the notional annual cost of a pump.
NumCalls	The total number of 999 calls received regarding this specific incident.
ResourceMobilisationId	The unique ID for each mobilized pump record.
Resource_Code	The code representing the station from which the pump was mobilized.
PerformanceReporting	Indicates the performance order of the pump arriving at the incident (e.g., 1st or 2nd).
DateAndTimeMobilised	The exact date and time (GMT) the mobilization order was given.
DateAndTimeMobile	The exact date and time (GMT) the mobilization actually started (the pump began moving).

DateAndTimeArrived	The exact date and time (GMT) the pump arrived at the incident location.
TurnoutTimeSeconds	The time taken in seconds from the mobilization order being given until the pump became mobile.
TravelTimeSeconds	The time taken in seconds from the pump becoming mobile until it reached the incident.
AttendanceTimeSeconds	The total time taken in seconds from the mobilization order being given until the pump reached the incident.
DateAndTimeLeft	The exact date and time (GMT) the pump left the incident location.
DateAndTimeReturned	The exact date and time (GMT) the pump returned to its station.
DeployedFromStation_Code	The specific code for the station from which the pump deployed.
DeployedFromStation_Name	The specific name of the station from which the pump deployed.
DeployedFromLocation	Specifies whether the pump deployed from its "Home Station" or an "Other Station".
PumpOrder	The sequential order of the pump arriving at the incident (categorized from 1 to 13).
PlusCode_Code	A short code for the mobilization type (Categories: Initial, rca, add).
PlusCode_Description	The description for the mobilization type (Categories: Initial Mobilisation, Running call, Addition to first attendance).

DelayCodeId	A short numerical code identifying the reason for a delayed mobilization.
DelayCode_Description	The textual description of the delayed mobilization reason (e.g., Traffic, Address incomplete, Appliance defect, Weather conditions).
Average_AttendanceTimeSeconds	(Target Label) The calculated average total time taken in seconds from the mobilization order being given until all pumps reached the incident.
Easting	The Easting spatial coordinate of the main station.
Northing	The Northing spatial coordinate of the main station.
distance	An engineered feature representing the physical distance in meters between the responding station and the incident location.

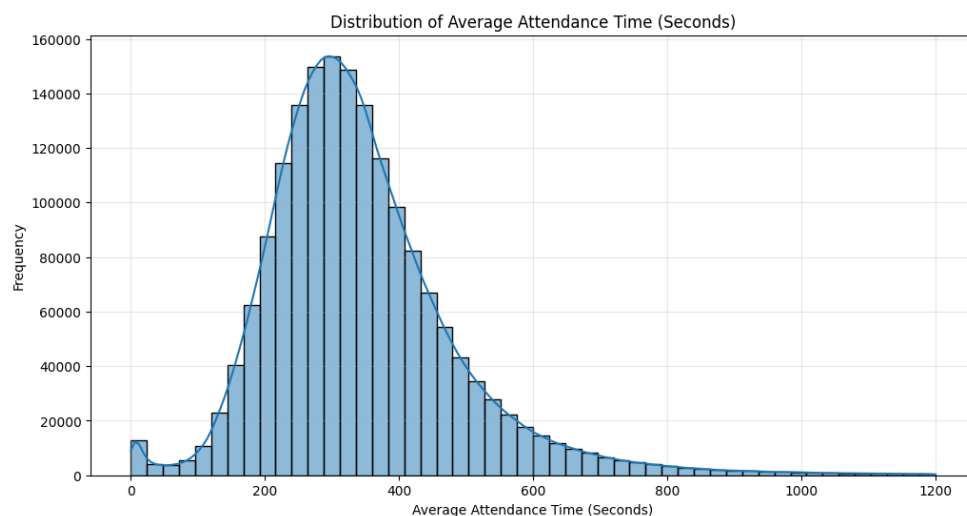
4. Data Exploration and Preprocessing

4.1. Target Variable Definition



The primary objective of the model is to estimate the **Average_AttendanceTimeSeconds**. This target variable represents the total average time elapsed from the moment a mobilization order is given until all deployed pumps reach the incident location.

- **Target Treatment:** To preserve the integrity of the predictive model and avoid synthetic bias, 834,804 records missing this target label were completely dropped.
- **Right skewness in the target:** the target label has high negative skewness with long tail from 800 to 1200 seconds which decreases any linear regression prediction.



4.2. Feature Selection and Dimensionality Reduction

To prevent data leakage, multicollinearity, and overfitting, a rigorous feature elimination process was applied. Out of the initial dataset, numerous features were excluded based on four main criteria:

1. **Direct Target Leakage:** Features that are direct components or highly correlated sub-metrics of the target label (e.g., FirstPumpArriving_AttendanceTime, TurnoutTimeSeconds, TravelTimeSeconds, AttendanceTimeSeconds).
2. **Redundancy and Duplication:** Geolocation and naming variables that duplicated existing data (e.g., Easting_rounded, Northing_rounded, Longitude, IncGeo_BoroughName, DeployedFromStation_Name).
3. **Administrative Identifiers:** Unique indexing strings and codes that hold no predictive value (e.g., IncidentNumber, ResourceMobilisationId).
4. **Post-Incident Logistics:** Variables recording events that occur after the attendance phase has concluded (e.g., DateAndTimeLeft, DateAndTimeReturned, Notional Cost (£)).

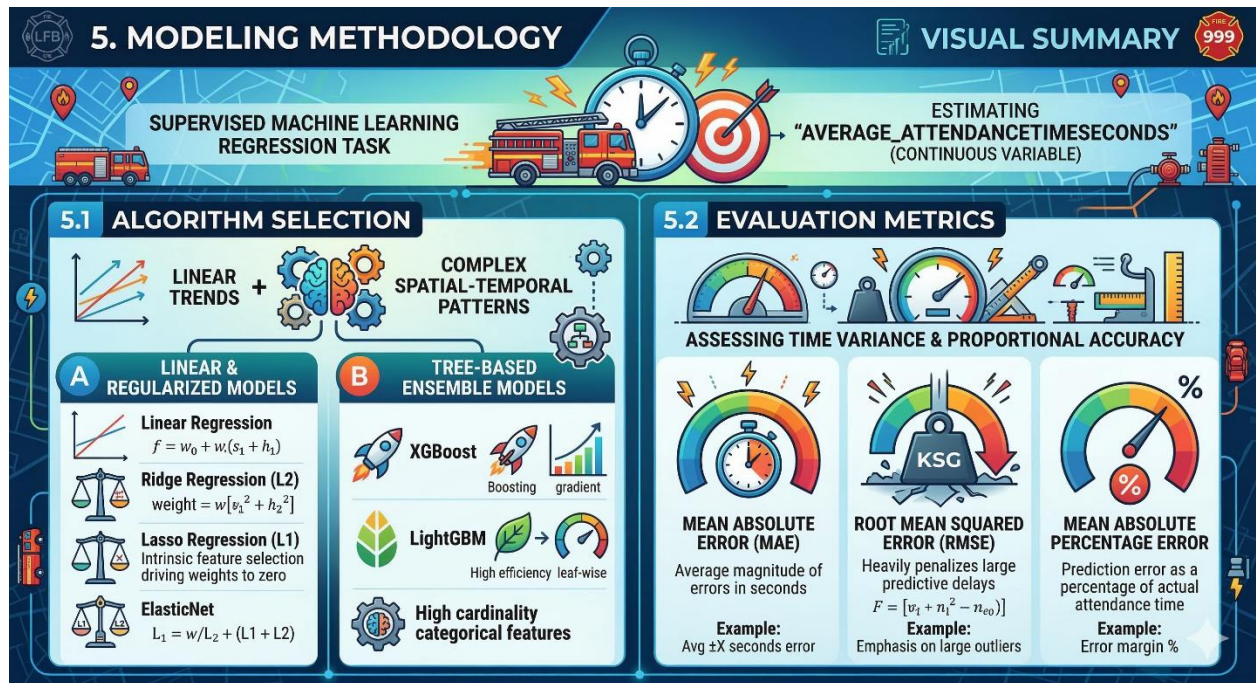
4.3. Retained Features and Imputation Strategy

The final set of features relies heavily on geographical routing constraints. Where missing values (NAs) were present, specific curing actions were applied based on the feature's statistical and operational nature to avoid introducing bias.

Feature Name	Preprocessing Action & NA Treatment
Temporal Data (CalYear, HourOfCall)	Kept as is; no missing values.
Incident Details (IncidentGroup, StopCodeDescription)	Kept. Rows with missing values in IncidentGroup were deleted as outliers.
SpecialServiceType	NAs were replaced with a specific "not special service" category.
Location Categories (PropertyCategory, PropertyType, AddressQualifier)	NAs were resolved automatically after deleting outlier rows in IncidentGroup.

Postcode_full	NAs were imputed using the broader Postcode_district code.
Spatial References (UPRN, USRN, IncGeo_WardCode, IncGeo_WardNameNew)	NAs were imputed using the statistical mode.
Spatial Coordinates (Easting_m, Northing_m)	NAs were imputed using the rounded Easting/Northing values.
Latitude	NAs were derived and filled using calculations from the rounded spatial values.
IncidentStationGround	Missing fields were specifically imputed with the "Addington" station.
FirstPumpArriving_DeployedFromStation	NAs were imputed using the statistical mode.
SecondPumpArriving_DeployedFromStation	NAs were filled by creating a "new category not used" to flag the missing status.
Logistics Metrics (NumStationsWithPumpsAttending, NumCalls, distance)	NAs were imputed using the statistical mean.
DelayCodeId	NAs were filled with 0, representing an assumption of "no delay".

5. Modeling Methodology



5.1. Algorithm Selection

Given the objective of estimating a continuous variable (Average_AttendanceTimeSeconds), this problem was framed as a supervised machine learning regression task. To capture both linear trends and complex spatial-temporal patterns, a diverse suite of algorithms was deployed:

- **Linear and Regularized Models:**
 - * **Linear Regression:** Utilized as the primary baseline algorithm to establish a fundamental linear relationship between the predictors and the target variable.
 - **Ridge Regression (L2 Regularization):** Applied to handle potential multicollinearity by penalizing the squared magnitude of coefficients.
 - **Lasso Regression (L1 Regularization):** Deployed to perform intrinsic feature selection, driving the weights of less impactful variables to zero.
 - **ElasticNet:** Utilized as a hybrid approach, combining L1 and L2 penalties to balance feature selection with coefficient stability.

- **Tree-Based Ensemble Models:** * **XGBoost (Extreme Gradient Boosting):** Selected for its robust capability to capture highly non-linear relationships and complex interactions between features.
 - **LightGBM (Light Gradient Boosting Machine):** Deployed for its high efficiency and speed on large datasets, utilizing leaf-wise tree growth to optimize predictive accuracy while managing high cardinality categorical features.

5.2. Evaluation Metrics

The final optimized models were evaluated using three primary statistical metrics to assess absolute time variance and proportional accuracy.

- **Mean Absolute Error (MAE):** The average magnitude of the errors in seconds.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

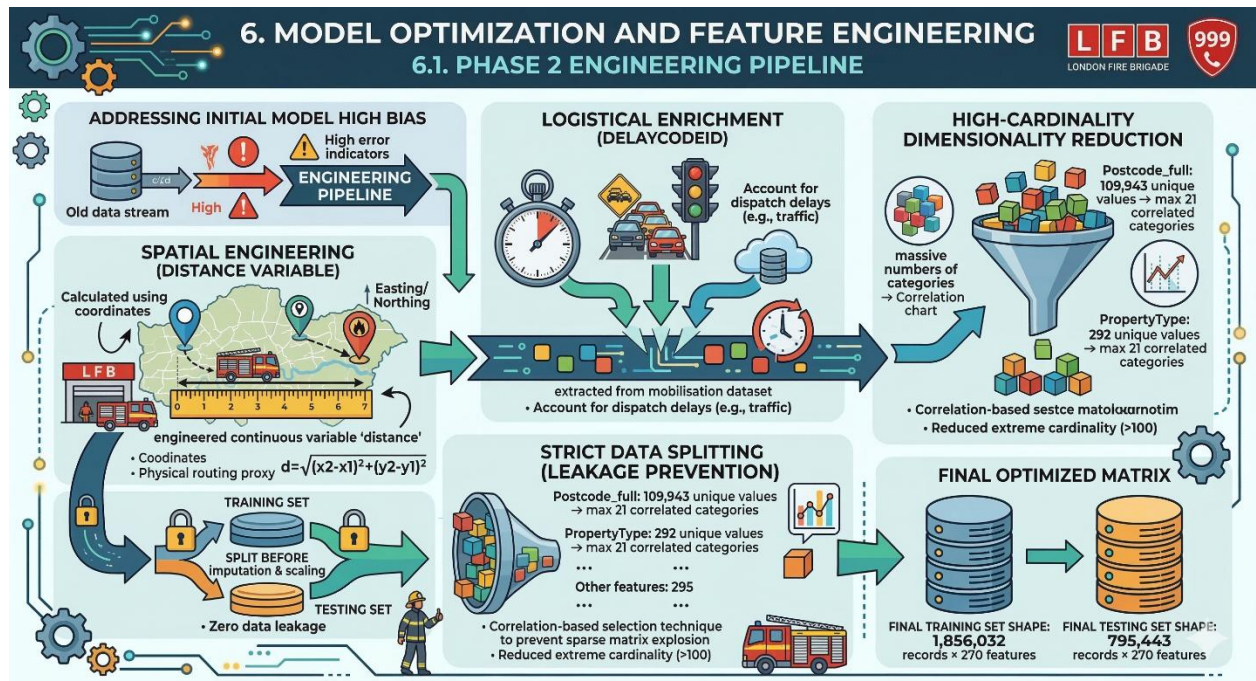
- **Root Mean Squared Error (RMSE):** Heavily penalizes large predictive delays.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Mean Absolute Percentage Error (MAPE):** Expresses the prediction error as a percentage of the actual attendance time.

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

6. Model Optimization and Feature Engineering



6.1. Phase 2 Engineering Pipeline

Following the high bias observed in the initial raw baseline models, a rigorous secondary phase of feature engineering and dimensionality reduction was executed:

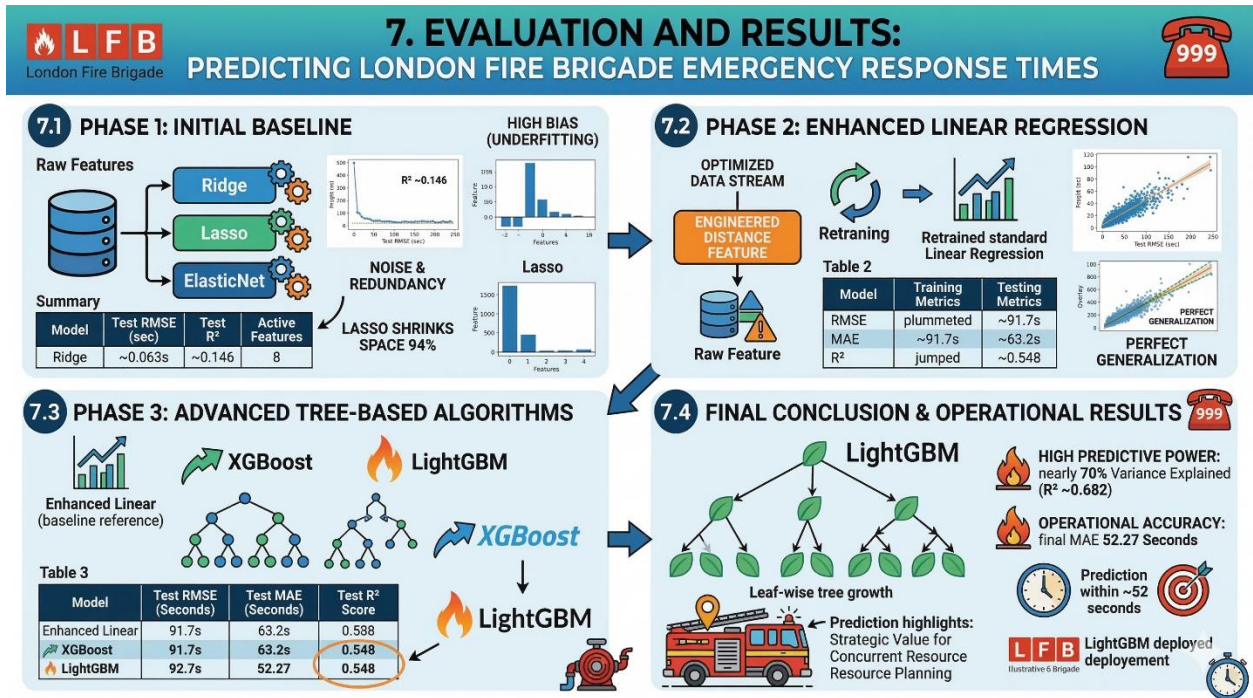
- Spatial Engineering:** A critical new continuous variable, distance, was engineered. This was calculated using the exact Easting and Northing spatial coordinates of both the incident location and the deploying LFB stations, providing the algorithms with a physical routing proxy.
- Logistical Enrichment:** The DelayCodeId feature was extracted from the mobilization dataset and integrated to account for physical dispatch delays (e.g., traffic).
- Strict Data Splitting (Leakage Prevention):** To ensure zero data leakage, the dataset was formally split into training and testing subsets *before* any missing value imputation or scaling occurred.
- High-Cardinality Dimensionality Reduction:** The dataset contained 15 categorical columns, 8 of which possessed extreme cardinality (exceeding 100 unique values), such as Postcode_full (109,943 unique values) and PropertyType (292 unique values). To prevent sparse matrix explosion during encoding, a correlation-based

selection technique was deployed. Each high-cardinality feature was reduced to a maximum of 21 highly correlated encoded categories.

This sophisticated preprocessing pipeline stabilized the dataset architecture, resulting in a highly optimized matrix:

- **Final Training Set Shape:** 1,856,032 records × 270 features
- **Final Testing Set Shape:** 795,443 records × 270 features

7. Evaluation and Results



7.1. Phase 1: Regularized Linear Models (Pre-Engineered Features)

Before introducing complex engineered spatial features or non-linear ensemble methods, an initial baseline was established using regularized linear regression techniques on the fully encoded, raw dataset consisting of 534 features.

Table 1: Initial Baseline Model Comparison (Raw Features)

Model	Train RMSE (sec)	Test RMSE (sec)	Train R ²	Test R ²	Active Features (Non-Zero)
Ridge	131.71	131.32	0.1466	0.1455	534 / 534 (100.0%)
Lasso	136.95	136.45	0.0772	0.0775	32 / 534 (6.0%)
ElasticNet	138.28	137.77	0.0593	0.0595	92 / 534 (17.2%)

- Phase 1 Insights:** The highest variance explained by any of the initial models was roughly 14.6% (Ridge Test R²: 0.1455), indicating high bias (underfitting). Furthermore, Lasso aggressively shrank the feature space, eliminating 94% of the variables with only a ~5-second penalty to the Test RMSE, proving the vast majority of initial encoded features were redundant noise.

7.2. Phase 2: Enhanced Linear Regression Results

With the newly engineered distance feature and the optimized 270-column feature space, the standard Linear Regression model was retrained

Table 2: Enhanced Linear Regression Performance

Metric	Training Data	Testing Data	Difference (Test - Train)
RMSE (Seconds)	92.05	91.71	-0.34
MAE (Seconds)	63.28	63.19	-0.09
R ² Score	0.5459	0.5477	+0.0018

- Phase 2 Insights:** The RMSE plummeted from roughly 131 seconds to 91.71 seconds, and the R² score jumped to 0.5477. The testing metrics are virtually identical to the training metrics, confirming perfect generalization to unseen data without overfitting.

7.3. Phase 3: Advanced Tree-Based Algorithms

To capture complex, multi-feature interactions, advanced gradient boosting frameworks—XGBoost and LightGBM—were deployed as the final tier of the modeling pipeline.

Table 3: Final Model Comparison (Gradient Boosting)

Model	Test RMSE (Seconds)	Test MAE (Seconds)	Test R ² Score
Linear Regression (Enhanced)	91.71	63.19	0.5477
XGBoost	76.98	52.31	0.6813
LightGBM	76.86	52.27	0.6823

7.4. Final Conclusion of Modeling Results

The deployment of LightGBM represents the apex of the predictive pipeline. By leveraging leaf-wise tree growth, LightGBM efficiently navigated the 270 features to establish the most accurate predictive bounds.

- **Operational Accuracy:** The LightGBM model achieved a final **Mean Absolute Error (MAE) of 52.27 seconds**. In the context of emergency services, predicting the arrival of fire brigade resources to within roughly 52 seconds provides immense strategic value for concurrent resource planning.
 - **High Predictive Power:** The final **R^2 score of 0.6823** indicates that nearly 70% of the complex variance in London's emergency response times has been successfully mapped and predicted by the algorithm.
-